| | School of Engineering & Technology | |
|---|---|---|
| | Department: SOET | Session: 25-26 Even |
| | Program: B. Tech CSE | Semester: IV |
| | Course Code: ENCS254 | Name: Yuvraj Singh<br>Roll No: 2401010225 |
| | Course Name: Database Management Systems Lab | Faculty: Ms. Mansi Kajal |

**Lab Sheet: 3**

**Objectives**

- To understand the use of COMMIT and ROLLBACK commands in transaction control.
- To use ORDER BY and GROUP BY clauses in SQL queries.
- To understand DCL commands (GRANT, REVOKE) and the use of SAVEPOINT.
- To understand and implement JOIN operations in SQL.

**Instructions**

Open your preferred Database Management System (DBMS) tool.

1. Create the required tables: TEACHER, SALES, DEPARTMENTS, and EMPLOYEES.
2. Insert suitable records into each table.
3. Execute the SQL commands for transaction control, privilege control, grouping, sorting, and joins.
4. Observe the output after each operation such as ROLLBACK, COMMIT, SAVEPOINT, GRANT, and REVOKE.

**DDL Questions (Data Definition Language)**

1. Create the **TEACHER** table with the following attributes:
   (Name, DeptNo, Date_of_Joining, DeptName, Location, Salary).
2. Create the **SALES** table with the following attributes:
   (Sales_No, Salesname, Branch, Salesamount, DOB).
3. Create the **DEPARTMENTS** table with the following attributes:
   (dept_no, dept_name, dept_location).
4. Create the **EMPLOYEES** table with the following attributes:
   (emp_id, emp_name, emp_salary, dept_no).

```
Database changed
mysql> CREATE TABLE TEACHER (
    ->     Name VARCHAR(30),
    ->     DeptNo INT,
    ->     Date_of_Joining DATE,
    ->     DeptName VARCHAR(30),
    ->     Location VARCHAR(30),
    ->     Salary INT
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE SALES (
    ->     Sales_No INT,
    ->     Salesname VARCHAR(30),
    ->     Branch VARCHAR(30),
    ->     Salesamount INT,
    ->     DOB DATE
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE TABLE DEPARTMENTS (
    ->     dept_no INT PRIMARY KEY,
    ->     dept_name VARCHAR(30),
    ->     dept_location VARCHAR(30)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE EMPLOYEES (
    ->     emp_id INT PRIMARY KEY,
    ->     emp_name VARCHAR(30),
    ->     emp_salary INT,
    ->     dept_no INT
    -> );
Query OK, 0 rows affected (0.03 sec)
```

**DML Questions (Data Manipulation Language)**

**From TEACHER Table**

1. Insert five records into the **TEACHER** table.
2. Give an increment of 25% salary to teachers working in the Mathematics department.
3. Give an increment of 15% salary to teachers working in the Commerce department.

**From SALES Table**

1. Insert five records into the **SALES** table.
2. Calculate the total sales amount in each branch.

3. Calculate the average sales amount in each branch.
4. Display the Salesname and DOB of salesmen born in the month of December with date format like *21-Dec-09*.
5. Display the Salesname and DOB of salesmen in alphabetical order of the month.

```
mysql> INSERT INTO TEACHER VALUES
    -> ('Ramesh',101,'2015-06-12','Mathematics','Delhi',50000),
    -> ('Sita',102,'2016-07-15','Commerce','Mumbai',45000),
    -> ('Anil',101,'2014-03-20','Mathematics','Delhi',52000),
    -> ('Priya',103,'2018-11-01','Science','Chennai',40000),
    -> ('Karan',102,'2017-09-10','Commerce','Mumbai',47000);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> UPDATE TEACHER
    -> SET Salary = Salary + (Salary * 0.15)
    -> WHERE DeptName = 'Commerce';
Query OK, 2 rows affected (0.01 sec)
Rows matched: 2  Changed: 2  Warnings: 0

mysql> INSERT INTO SALES VALUES
    -> (1,'Raj','Delhi',20000,'1990-12-21'),
    -> (2,'Amit','Mumbai',15000,'1992-05-10'),
    -> (3,'Neha','Delhi',18000,'1989-12-11'),
    -> (4,'John','Chennai',22000,'1995-08-19'),
    -> (5,'Sara','Mumbai',17000,'1991-12-25');
Query OK, 5 rows affected (0.00 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT Branch, SUM(Salesamount) AS Total_Sales
    -> FROM SALES
    -> GROUP BY Branch;
+---------+-------------+
| Branch  | Total_Sales |
+---------+-------------+
| Delhi   |       38000 |
| Mumbai  |       32000 |
| Chennai |       22000 |
+---------+-------------+
3 rows in set (0.01 sec)

mysql> SELECT Branch, AVG(Salesamount) AS Average_Sales
    -> FROM SALES
    -> GROUP BY Branch;
+---------+--------------+
| Branch  | Average_Sales |
+---------+--------------+
| Delhi   |    19000.0000 |
| Mumbai  |    16000.0000 |
| Chennai |    22000.0000 |
+---------+--------------+
3 rows in set (0.00 sec)

mysql> SELECT Salesname,
    -> DATE_FORMAT(DOB,'%d-%b-%y') AS DOB
    -> FROM SALES
    -> WHERE MONTH(DOB) = 12;
```

```
    -> WHERE MONTH(DOB) = 12;
+-----------+-----------+
| Salesname | DOB       |
+-----------+-----------+
| Raj       | 21-Dec-90 |
| Neha      | 11-Dec-89 |
| Sara      | 25-Dec-91 |
+-----------+-----------+
3 rows in set (0.01 sec)

mysql> SELECT Salesname,
    -> DATE_FORMAT(DOB,'%d-%b-%y') AS DOB
    -> FROM SALES
    -> ORDER BY MONTHNAME(DOB);
+-----------+-----------+
| Salesname | DOB       |
+-----------+-----------+
| John      | 19-Aug-95 |
| Raj       | 21-Dec-90 |
| Neha      | 11-Dec-89 |
| Sara      | 25-Dec-91 |
| Amit      | 10-May-92 |
+-----------+-----------+
5 rows in set (0.01 sec)
```

**From JOIN Operations**

1. Display employee details and department details where the department numbers are the same in both tables.
2. Display the employee name and department name using a LEFT OUTER JOIN.
3. Display the employee name and department name using a RIGHT OUTER JOIN.
4. Display the details of employees who draw a salary greater than the average salary.

```
mysql> SELECT *
    -> FROM EMPLOYEES E
    -> JOIN DEPARTMENTS D
    -> ON E.dept_no = D.dept_no;
+--------+----------+------------+---------+---------+-----------+---------------+
| emp_id | emp_name | emp_salary | dept_no | dept_no | dept_name | dept_location |
+--------+----------+------------+---------+---------+-----------+---------------+
|      1 | Rohit    |      50000 |      10 |      10 | HR        | Delhi         |
|      2 | Sneha    |      60000 |      20 |      20 | Finance   | Mumbai        |
|      3 | Arjun    |      55000 |      30 |      30 | IT        | Chennai       |
```

```
mysql> SELECT E.emp_name, D.dept_name
    -> FROM EMPLOYEES E
    -> LEFT JOIN DEPARTMENTS D
    -> ON E.dept_no = D.dept_no;
+----------+-----------+
| emp_name | dept_name |
+----------+-----------+
| Rohit    | HR        |
| Sneha    | Finance   |
| Arjun    | IT        |
| Meena    | HR        |
+----------+-----------+
4 rows in set (0.00 sec)

mysql> SELECT E.emp_name, D.dept_name
    -> FROM EMPLOYEES E
    -> RIGHT JOIN DEPARTMENTS D
    -> ON E.dept_no = D.dept_no;
+----------+-----------+
| emp_name | dept_name |
+----------+-----------+
| Meena    | HR        |
| Rohit    | HR        |
| Sneha    | Finance   |
| Arjun    | IT        |
+----------+-----------+
4 rows in set (0.00 sec)

mysql> SELECT *
    -> FROM EMPLOYEES
    -> WHERE emp_salary >
    -> (SELECT AVG(emp_salary) FROM EMPLOYEES);
+--------+----------+------------+---------+
| emp_id | emp_name | emp_salary | dept_no |
+--------+----------+------------+---------+
|      2 | Sneha    |      60000 |      20 |
|      3 | Arjun    |      55000 |      30 |
+--------+----------+------------+---------+
2 rows in set (0.00 sec)
```

**TCL Questions (Transaction Control Language)**

1. Perform the **ROLLBACK** command on the TEACHER table.
2. Perform the **COMMIT** command on the TEACHER table.
3. Write a query to implement a **SAVEPOINT**.

**DCL Questions (Data Control Language)**

1. Write a query to **grant all privileges** on the EMPLOYEES table to DEPARTMENTS.
2. Write a query to **grant selected privileges** on the EMPLOYEES table to DEPARTMENTS.

3. Write a query to **revoke all privileges** on the EMPLOYEES table from DEPARTMENTS.
4. Write a query to **revoke selected privileges** on the EMPLOYEES table from DEPARTMENTS.

```
mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> SAVEPOINT sp1;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> grant all privileges
    -> on DBMS_Vandana.EMPLOYEES
    -> to'testuser'@''localhost';
    '>
    '> grant all privileges
    '> to 'testuser'@'localhost';
    '> grant select,insert
    '> on DBMS_Vandana.EMPLOYEES
    '> to'testuser'@'localhost';
    '> revoke all privileges
    '> on DBMS_Vandana.EMPLOYEES
    '> from 'testuser'@'localhost';
    '> revoke select,insert
    '> on DBMS_Vandana.EMPLOYEES
    '> from'testuser'@'localhost';
```